

ANUBHAV SAINI

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EDUCATION

Bachelor of Applied Science in Engineering Physics

University of British Columbia – Specialization in Electrical Engineering and Robotics

Expected May 2027

Vancouver, BC

EXPERIENCE

Electronics Hardware Intern

January 2026 – August 2026

Kardium Inc.

Burnaby, BC

- Designed the **2nd generation of Kardium's highest-volume production PCB** – a **6-layer** board carrying both high-voltage and low-voltage signals – **increasing its HIPOT withstand strength** for medical HV safety compliance.
- Re-defined PCB design rules and fabrication drawings and updated the **6-layer** design to a new vendor's specifications while maintaining **IPC Class 3** requirements, **cutting production costs** on the safety-critical, high-volume board.
- Executed layout for Kardium's next generation of **medical-grade flex PCBs**, routing **1500V+ signals** within strict safety clearances to deliver short-term cost reduction and rapid drop-in integration into the current production system.
- Spearheaded **end-to-end validation of a 12-layer medical-grade PCB**: authored and executed **25+ tests across 5 diagnostic suites** (Power, Signal Integrity, Thermal, ADC, Isolation) for **100% test coverage**, building custom **10x/100x coaxial probes** to capture high-fidelity signal-integrity and isolation measurements across the board.
- Debugged multiple **14-layer boards** within a **10+ board system** via high-speed signal analysis and precision rework.

Hardware Engineering Intern

May 2025 – December 2025

Arlo Technologies

Vancouver, BC

- Designed a cost-effective **Power-ORing Switch** (integrated FW + auxiliary HW control) with **98% efficiency** to charge Arlo's camera from the best available power source; applied transistor/diode array logic to create a discrete HV input system.
- Led R&D of a new solar-powered security camera with an embedded solar panel**, defined system architecture (configuration of multiple power inputs, finding Boost/charger/fuel gauge ICs), conducted competitive benchmarking, and validated design through comprehensive voltage, power, efficiency, and battery charge/discharge cycle testing.
- Investigated and eliminated PIR noise induced by RF radiation from 2.4 GHz Wifi, **cutting yield loss from 30% to less than 1%** and ensuring reliable motion detection during all wireless camera operating states (idle, streaming, recording).
- Increased hardware test coverage for PIR and Ambient Light Sensors **from 60% to 100%** to build a coexistence test suite.

Electrical Engineering Intern

September 2024 – December 2024

Sarcomere Dynamics Inc.

Vancouver, BC

- Expedited firmware for an in-house fingertip force sensor **5-10 times cheaper** than on-shelf solutions, reconstructing Normal Force and Shear Direction from raw tri-axis magnetometer (MLX90393) data with **more than 90% accuracy** in real time.
- Devised a computationally light calibration using pseudo-inverse matrix mapping, verified via K-Means clustering.
- Engineered a **Motor Driver PCB** in **Altium** around an **STM32F412** – SPI to the BLDC motor driver, I2C magnetometer feedback for rotor positioning, CAN for multi-board scalability, and **3.3V** on-board linear regulation from a single 5V input.

Product Engineering Intern

May 2024 – August 2024

Microchip Technology

Burnaby, BC

- Programmed Python test scripts and custom firmware to characterize the SERDES I3C Pad, executing **8 AC and 5 DC tests** across **5 voltage/temperature (VT) corners** with per-test BERT/DCSU configuration and pass/fail yield checks.
- Formulated and deployed a Multi-threaded testing procedure to **reduce testing time by more than 40%** – inverted the test loop to hold each VT corner fixed while threaded Python maintained uninterrupted voltage and temperature control.
- Reduced power consumption by 15%** via systematic voltage/temperature regularization during characterization.

PERSONAL PROJECTS

Solar Express – DC Power Optimizer | *Power Electronics, SPICE, Control Loop Design, Python, MPPT*

- Architected a **600W** panel-level DC-DC power optimizer around a **4-switch buck-boost** with **100V GaN FETs at 250kHz**, spanning a 10-80V input / 5-80V output envelope at **15A**; defined the full spec from PV-panel and string-inverter constraints.
- Validated the power stage via automated NGSpice switching simulations across a **24-point $V_{in} \times V_{out}$ grid**, achieving **99.5% pass-through** and **99.2% boost efficiency**; verified NEC 690.12 rapid shutdown to **under 1V in 1.1s** (2s spec).
- Performed small-signal plant identification on cycle-averaged SPICE models – characterized the boost-mode **RHP zero (3.9kHz)** and built a **Type-II compensator** with $\geq 94^\circ$ **phase margin**, $\geq 10\text{dB}$ **gain margin**, and **0.12%** step deviation.
- Optimized component selection via parametric loss-budget sweeps ($R_{DS(on)}$ 3-22m Ω , f_{sw} 100kHz-1MHz, L 3.3-22 μH) to lock a **6.8 μH / 3.2m Ω / 250kHz** design point; built a **Python + NGSpice** pipeline validated within $\pm 0.5\%$ of an analytic model.

Automated Robot Race-Car | *C/C++, Embedded Systems, Analog Circuit Design, PID*

- Built circuitry integrating sensors (IMU, magnetometer, self-made rotary encoder for wheel-spin position mapping) with an STM32, applying RC filtering to cut motor noise and fuse real-time orientation and location data with **over 85% accuracy**.
- Applied PID control to autonomously track a preset optimal path and developed multi-sensor recalibration routines to re-localize the car after **1.5-foot ramp jumps** and obstacle encounters, **achieving the fastest lap time by 4 seconds**.

TECHNICAL SKILLS

PCB Design: Altium Designer, Schematic Capture, 6-14 Layer & Flex Layout, High-Voltage Design (HIPOT, IPC Class 3), DFM

Power & Validation: Power Electronics (DC-DC), Signal Integrity, SPICE, Oscilloscopes & Probing, Soldering & Rework

Embedded & Software: STM32, SPI, I2C, UART, CAN, Python, C/C++, MATLAB, TensorFlow, OpenCV, ROS, Linux